

# PROFESSIONAL PORTFOLIO



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**BUILT ON PRECISION.  
DRIVEN BY QUALITY.**

**ENGR. RON MAR B. PANGILINAN**

QA/QC CIVIL ENGINEER | PLANNING ENGINEER | CONSTRUCTION ESTIMATOR | QUANTITY TAKEOFF SPECIALIST



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## PROFILE SUMMARY:

Registered Civil Engineer and Construction Estimator with hands-on experience in **quantity takeoff, cost estimating, structural/civil engineering, and QA/QC**. Skilled in interpreting architectural, structural, civil, and MEP drawings and developing detailed quantity takeoffs, **BOQ/BOM**, and cost estimates.

Experienced in construction planning, project coordination, quality control, and technical documentation within a multinational solar power project. Proficient in **PlanSwift, Bluebeam Revu, Microsoft Excel, AutoCAD, and Primavera P6**.

Focused on delivering accurate quantities, practical solutions, quality workmanship, and well-organized construction documentation—from planning and design through execution.

## CORE STRENGTHS

QUANTITY TAKEOFF • COST ESTIMATING • BOQ/BOM • QA/QC • DRAWING INTERPRETATION  
CONSTRUCTION PLANNING • PROJECT COORDINATION • TECHNICAL DOCUMENTATION

## SOFTWARE & TECHNICAL SKILLS 02

### PLANSWIFT 11

Materials & Labor Quantity Takeoff | Cost Estimating Reports | Custom Assemblies & Automated Takeoff | Structural Concrete & Rebar | Architectural Works | Roofing Works | Multi-trade Estimating



### BLUEBEAM REVU 21

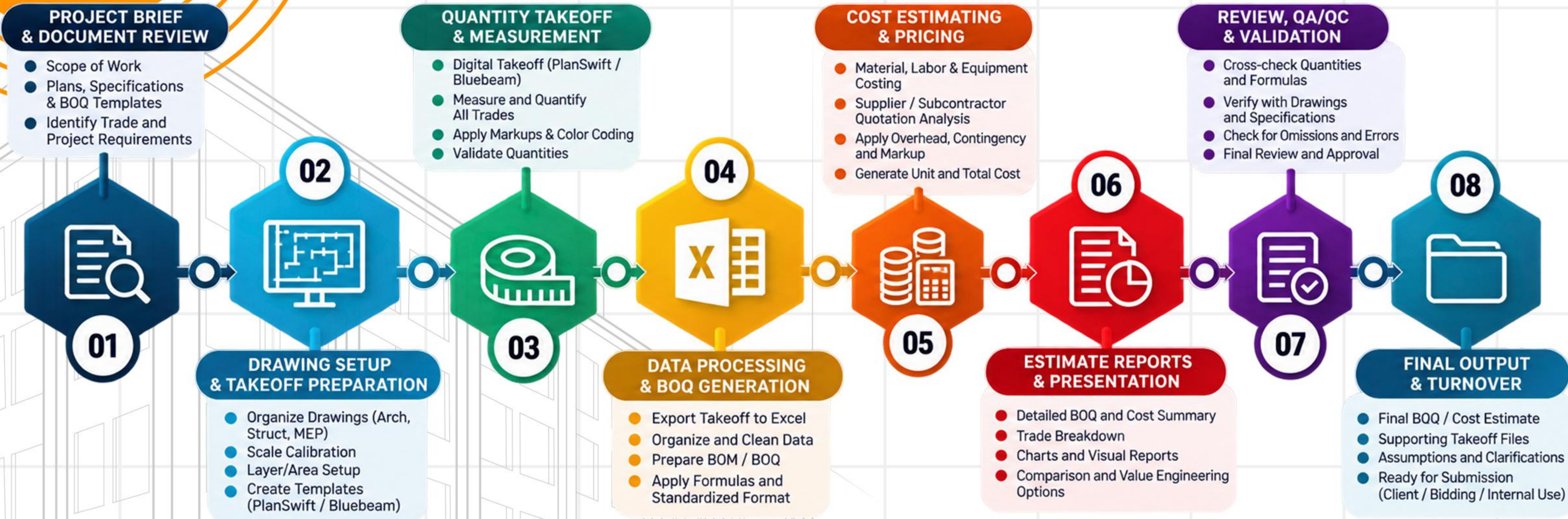
Material & Labor Quantity Takeoff | PDF Measurement & Markups | Construction Document Review | Architectural, Concrete & MEP Estimating

## ENGINEERING & DESIGN SOFTWARE



## REMOTE WORK & COLLABORATION TOOLS





04 SAMPLE ESTIMATING PROJECTS

1. PROPOSED TWO-STOREY RESIDENCE – 178.70 sq.m (1,923 sq.ft.)

Estimated Construction Cost: USD 96,586.10

- Item Summary:
- General Requirements
  - Siteworks
  - Civil / Structural Works
  - Architectural Works
  - Plumbing Works
  - Electrical Works
  - Tinsmithry / Roofing Works
  - Metal Works
  - Cabinetry and other Finishing Works



Note: Drawings and project information are presented for training and professional portfolio purposes. Actual project records and confidential information are not reproduced.

SAMPLE BILL OF MATERIALS

| BILL OF MATERIALS |                                               |          |       |                 |             |
|-------------------|-----------------------------------------------|----------|-------|-----------------|-------------|
| Item No.          | Item Description                              | Quantity | Unit  | Unit Price, Php | Amount, Php |
| III.              | CIVIL/STRUCTURAL WORKS                        |          |       |                 |             |
| III.i.            | CONCRETING/MASONRY WORKS/REINFORCEMENTS       |          |       |                 |             |
|                   | RC Footings (3000 PSI)                        | 6.77     | cu.m  |                 |             |
|                   | • Cement                                      | 51.00    | bgs   | 250.00          | 12,750.00   |
|                   | • Sand                                        | 4.00     | cu.m  | 1,000.00        | 4,000.00    |
|                   | • Gravel 3/4"                                 | 7.00     | cu.m  | 1,450.00        | 10,150.00   |
|                   | • 16 mmø def bar (6 meters)                   | 48.00    | pcs   | 480.00          | 23,040.00   |
|                   | • 10 mmø def bar (6 meters)                   | 8.00     | pcs   | 230.00          | 1,840.00    |
|                   | • GI Tie Wire # 16                            | 5.00     | kgs   | 85.00           | 425.00      |
|                   | RC Columns (3000 PSI)                         | 9.43     | cu.m  |                 |             |
|                   | • Cement                                      | 71.00    | bgs   | 250.00          | 17,750.00   |
|                   | • Sand                                        | 5.00     | cu.m  | 1,000.00        | 5,000.00    |
|                   | • Gravel 3/4"                                 | 10.00    | cu.m  | 1,450.00        | 14,500.00   |
|                   | • 16 mmø def bar (6 meters)                   | 140.00   | pcs   | 480.00          | 67,200.00   |
|                   | • 10 mmø def bar (6 meters)                   | 139.00   | pcs   | 230.00          | 31,970.00   |
|                   | • GI Tie Wire # 16                            | 19.00    | kgs   | 85.00           | 1,615.00    |
|                   | Tie Beams (3000 PSI)                          | 3.49     | cu.m  |                 |             |
|                   | • Cement                                      | 27.00    | bgs   | 250.00          | 6,750.00    |
|                   | • Sand                                        | 2.00     | cu.m  | 1,000.00        | 2,000.00    |
|                   | • Gravel 3/4"                                 | 4.00     | cu.m  | 1,450.00        | 5,800.00    |
|                   | • 12 mmø def bar (6 meters)                   | 57.00    | pcs   | 290.00          | 16,530.00   |
|                   | • 10 mmø def bar (6 meters)                   | 49.00    | pcs   | 230.00          | 11,270.00   |
|                   | • GI Tie Wire # 16                            | 5.00     | kgs   | 85.00           | 425.00      |
|                   | Slab on Grade (3000 PSI)                      | 11.63    | cu.m  |                 |             |
|                   | • Cement                                      | 88.00    | bgs   | 250.00          | 22,000.00   |
|                   | • Sand                                        | 6.00     | cu.m  | 1,000.00        | 6,000.00    |
|                   | • Gravel 3/4"                                 | 12.00    | cu.m  | 1,450.00        | 17,400.00   |
|                   | • 10 mmø def bar (6 meters)                   | 127.00   | pcs   | 230.00          | 29,210.00   |
|                   | • GI Tie Wire # 16                            | 5.00     | kgs   | 85.00           | 425.00      |
|                   | Masonry Works                                 | 245.16   | sq.m  |                 |             |
|                   | • Concrete hollow block - 4"                  | 944.00   | pcs   | 15.00           | 14,160.00   |
|                   | • Concrete hollow block - 6"                  | 2428.00  | pcs   | 18.00           | 43,704.00   |
|                   | • Cement                                      | 227.00   | bgs   | 250.00          | 56,750.00   |
|                   | • Sand                                        | 19.00    | cu.m  | 1,000.00        | 19,000.00   |
|                   | • 10 mmø def bar (6 meters)                   | 189.00   | pcs   | 230.00          | 43,470.00   |
|                   | • GI Tie Wire # 16                            | 7.00     | kgs   | 85.00           | 595.00      |
| III.ii.           | FORMWORKS                                     | 170.26   | sqm   |                 |             |
|                   | • Phenolic Board (4 x 8 x 1/2)                | 81.00    | pcs   | 700.00          | 56,700.00   |
|                   | • CWN Assorted Nail                           | 117.00   | kgs   | 80.00           | 9,360.00    |
|                   | • Lumber for formworks ( 2" x 3" and 2" x 2") | 5197.00  | bd.ft | 60.00           | 311,820.00  |
|                   | • Standard H-Frame Set (1.2 x 1.7)            | 4.00     | set   | 450.00          | 3,600.00    |
|                   | • Platforms/Catwalk                           | 8.00     | pcs   | 350.00          | 5,600.00    |
|                   | • Consumables/miscellaneous                   | 1.00     | lot   | 5%              | 19,354.00   |

| BILL OF MATERIALS |                                                                           |          |       |                 |             |
|-------------------|---------------------------------------------------------------------------|----------|-------|-----------------|-------------|
| Item No.          | Item Description                                                          | Quantity | Unit  | Unit Price, Php | Amount, Php |
| IV.               | ARCHITECTURAL WORKS                                                       |          |       |                 |             |
| IV.i              | CEILING WORKS                                                             | 202.35   | sqm   |                 |             |
|                   | • Metal Furring (JEAMaxx G.24)                                            | 63.00    | pcs   | 130.00          | 8,190.00    |
|                   | • Carrying Channel (JEAMaxx G.24)                                         | 59.00    | pcs   | 140.00          | 8,260.00    |
|                   | • Double U-Clip (JEAMAx)                                                  | 391.00   | pcs   | 5.00            | 1,955.00    |
|                   | • Wall Angle (JEAMaxx G.24)                                               | 148.00   | pcs   | 80.00           | 11,840.00   |
|                   | • Blind rivets 1/8" dia. X 1/2" (1000 PCS PER BOX)                        | 4.00     | boxes | 450.00          | 1,800.00    |
|                   | • Gypsum Board (9mm)                                                      | 53.00    | pcs   | 750.00          | 39,750.00   |
|                   | • Gypsum Mesh Tape                                                        | 6.00     | pcs   | 120.00          | 720.00      |
|                   | • Metal Spandrel (0.15 x 0.8)                                             | 603.00   | pcs   | 75.00           | 45,225.00   |
|                   | • black screw                                                             | 1866.00  | pcs   | 1.00            | 1,866.00    |
|                   | • Consumables/miscellaneous                                               | 1.00     | lot   | 10%             | 11,960.60   |
|                   |                                                                           |          |       | Sub Total =     | 131,566.60  |
|                   |                                                                           |          |       | Labor Cost =    | 52,626.64   |
| IV.ii             | DOORS & GLASSES                                                           | 43.47    | sqm   |                 |             |
|                   | GLASSES                                                                   |          |       |                 |             |
|                   | • GLASS ENCLOSURE                                                         | 2.00     | set   | 15,000.00       | 30,000.00   |
|                   | • Sliding Door (2-panel 1.6 x 2.1 with 1/4" Bronze glass & analok finish) | 2.00     | set   | 20,000.00       | 40,000.00   |
|                   | • Fixed Window With 1/4" Glass (1.2 x 2.1 )                               | 1.00     | set   | 12,000.00       | 12,000.00   |
|                   | • Awning Windows & Fixed (0.6 x 0.6)                                      | 2.00     | set   | 3,000.00        | 6,000.00    |
|                   | • Exterior Windows (1.0 x 1.2 )                                           | 7.00     | set   | 5,000.00        | 35,000.00   |
|                   | • Exterior Windows (1.0 x 1.0)                                            | 3.00     | set   | 5,000.00        | 15,000.00   |
|                   | • Fixed Window With Glass (0.6 x 5.4 )                                    | 3.00     | set   | 18,000.00       | 54,000.00   |
|                   | DOORS                                                                     |          |       |                 |             |
|                   | • Matwood Solid Panel Doors (1.2 x 2.1 ) WITH ACCESSORIES                 | 1.00     | set   | 20,000.00       | 20,000.00   |
|                   | • Matwood Panel Type Doors (0.8 x 2.1)                                    | 3.00     | set   | 18,000.00       | 54,000.00   |
|                   | • Matwood Panel Type Doors (0.7 x 2.1)                                    | 1.00     | set   | 15,000.00       | 15,000.00   |
|                   | • CR Door - Painted Flush Door (0.8 x 2.1)                                | 2.00     | set   | 10,000.00       | 20,000.00   |
|                   |                                                                           |          |       | Sub Total =     | 301,000.00  |
|                   |                                                                           |          |       | Labor Cost =    | 120,400.00  |
| IV.iii            | TILE WORKS (Supply and installation of wall and floor tiles)              | 175.14   | sqm   |                 |             |
|                   | • Porcelain (0.60x0.60)                                                   | 145.00   | pcs   | 350.00          | 50,750.00   |
|                   | • Anti-Slip Ceramic (0.40x0.40)                                           | 130.00   | pcs   | 70.00           | 9,100.00    |
|                   | • Anti-Slip Ceramic (0.30x0.30)                                           | 62.00    | pcs   | 40.00           | 2,480.00    |
|                   | • Porcelain (0.20x0.60)                                                   | 145.00   | pcs   | 200.00          | 29,000.00   |
|                   | • Marble Tile (0.30x0.30)                                                 | 36.00    | pcs   | 100.00          | 3,600.00    |
|                   | • Wood-look Tile (0.15x0.60)                                              | 414.00   | pcs   | 120.00          | 49,680.00   |
|                   | • Wood-look Tile (0.20x1.0)                                               | 252.00   | pcs   | 170.00          | 42,840.00   |
|                   | • Herringbone Brick (0.20x0.10)                                           | 481.00   | pcs   | 40.00           | 19,240.00   |
|                   | • Tile Adhesive                                                           | 16.00    | bags  | 350.00          | 5,600.00    |
|                   | • Tile Adhesive (Heavy Duty)                                              | 6.00     | bags  | 530.00          | 3,180.00    |
|                   | • Cement                                                                  | 56.00    | bgs   | 250.00          | 14,000.00   |
|                   | • Sand                                                                    | 5.00     | cu.m  | 1,000.00        | 5,000.00    |
|                   | • Tile Grout (2kg)                                                        | 46.00    | bags  | 113.00          | 5,198.00    |
|                   | • tile spacers                                                            | 4736.00  | pcs   | 2.00            | 9,472.00    |
|                   | • Consumables/miscellaneous                                               | 1.00     | lot   | 5%              | 12,457.00   |
|                   |                                                                           |          |       | Sub Total =     | 261,597.00  |
|                   |                                                                           |          |       | Labor Cost =    | 104,638.80  |

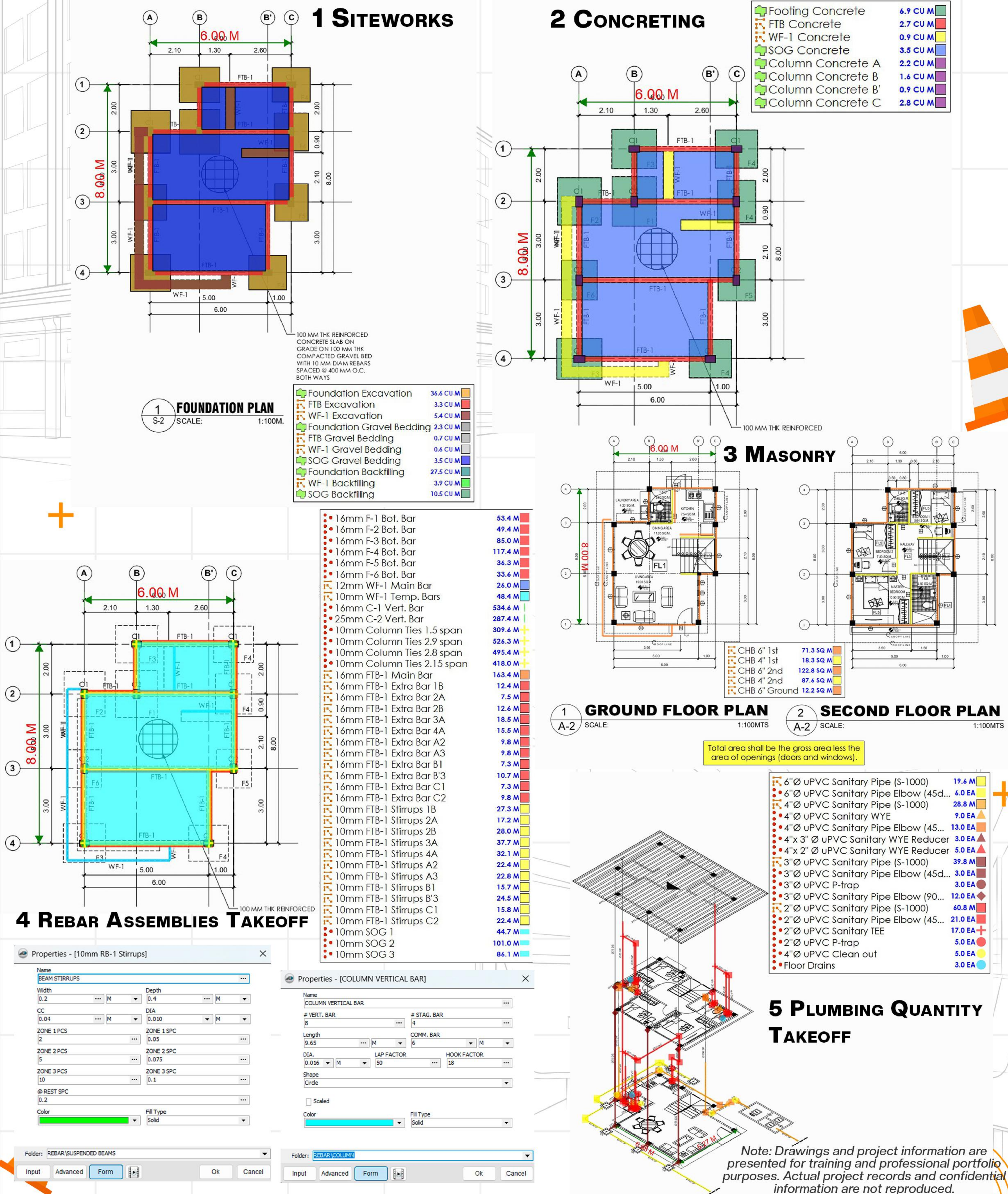
# 04 SAMPLE ESTIMATING PROJECTS

## 2. PROPOSED TWO-STOREY RESIDENTIAL BUILDING – 97.50 SQ.M. (1,049 SQ.FT.)

Estimated Construction Cost: USD 81,712.43

Item Summary: General Requirements | Siteworks | Civil / Structural Works | Architectural Works  
Plumbing Works | Tinsmithry / Roofing Works

### SAMPLE PLANSWIFT QUANTITY TAKEOFF



# 04 SAMPLE ESTIMATING PROJECTS

## 3. PROPOSED TWO-STOREY RESIDENTIAL BUILDING – 198.04 sq.M. (2,132 sq.FT.)

**Estimated Construction Cost: USD 51,702.03**  
**Item Summary: Siteworks / Concrete Works**  
**Reinforcement Works / Formworks**  
**Ceiling Works / Tile Works**



### RMP ENGINEERING SERVICES

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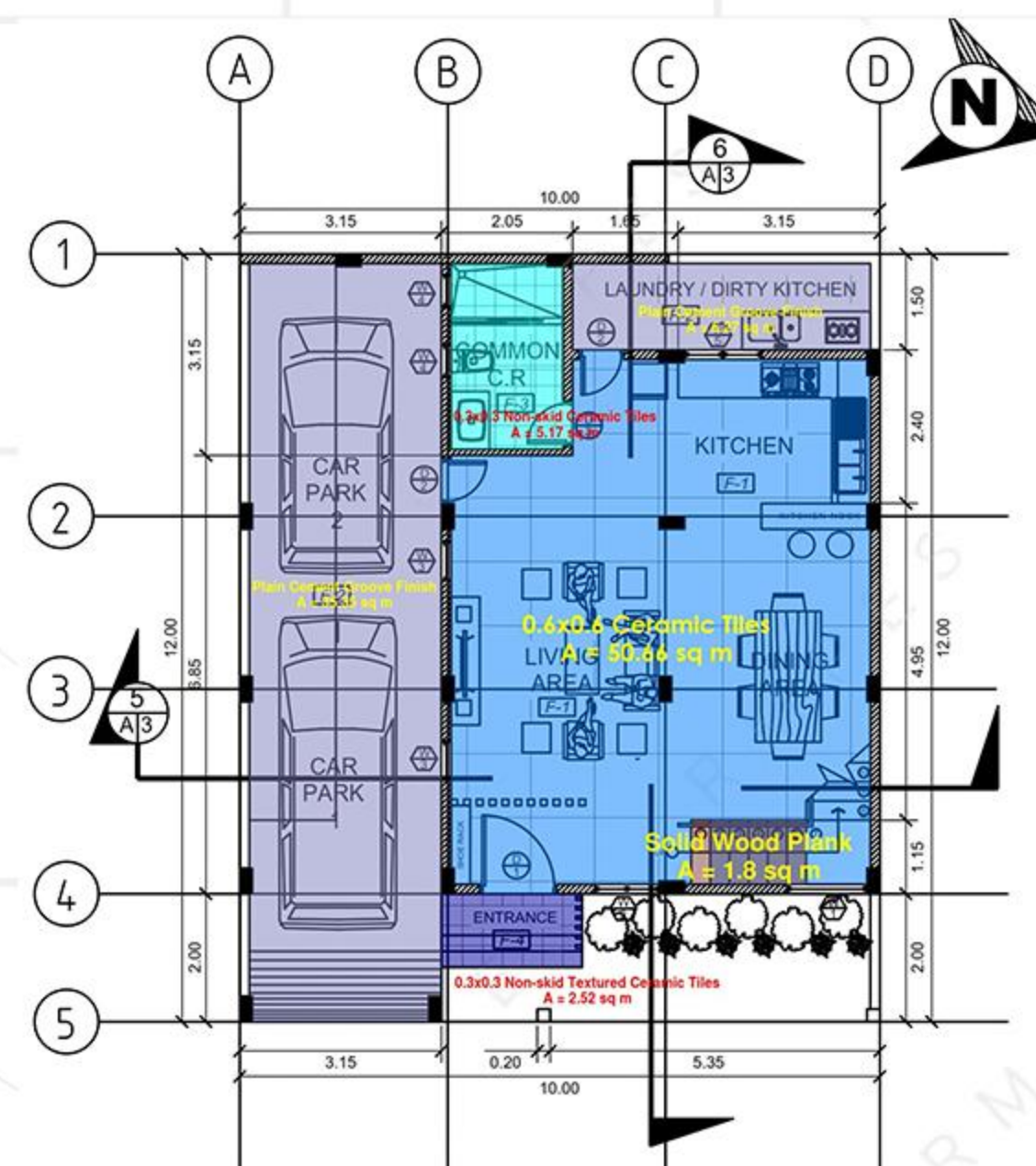
#### CONSTRUCTION QUANTITY TAKE-OFF

| Name                                    | Description | Qty     | Units |
|-----------------------------------------|-------------|---------|-------|
| <b>SITWORKS</b>                         |             |         |       |
| TOTAL EXCAVATION                        |             | 38      | CU M  |
| TOTAL GRAVEL BEDDING                    |             | 9.7     | CU M  |
| TOTAL BACKFILLING                       |             | 72.82   | CU M  |
| <b>CONCRETING</b>                       |             |         |       |
| Lean Concrete                           |             | 1.89    | CU M  |
| Footing Concrete                        |             | 5.71    | CU M  |
| FTB Concrete                            |             | 3.51    | CU M  |
| SOG Concrete                            |             | 9.35    | CU M  |
| Column Concrete                         |             | 12.02   | CU M  |
| Suspended Beam Concrete                 |             | 6.08    | CU M  |
| S-1 Concrete                            |             | 6.37    | CU M  |
| RB-1 Concrete                           |             | 2.65    | CU M  |
| <b>REBAR</b>                            |             |         |       |
| <b>FOOTING</b>                          |             |         |       |
| 16mm Dia. Footing Rebar                 |             | 245.76  | M     |
| <b>COLUMN</b>                           |             |         |       |
| 16mm Column Rebar                       |             | 526.88  | M     |
| 12mm Column Rebar                       |             | 317.6   | M     |
| 10mm Column Rebar                       |             | 1855.75 | M     |
| <b>FTB-1</b>                            |             |         |       |
| 16mm FTB-1 Rebar                        |             | 313.1   | M     |
| 10mm FTB-1 Rebar                        |             | 448.14  | M     |
| <b>SOG</b>                              |             |         |       |
| 10mm SOG Rebar                          |             | 525.12  | M     |
| <b>SUSPENDED BEAMS</b>                  |             |         |       |
| 16mm Suspended Beams                    |             | 532.85  | M     |
| 10mm Suspended Beams                    |             | 773.68  | M     |
| <b>SUSPENDED SLABS</b>                  |             |         |       |
| 10mm S-1 Rebar                          |             | 1281.43 | M     |
| <b>ROOF BEAMS</b>                       |             |         |       |
| 12mm RB-1 Main Rebar                    |             | 238.76  | M     |
| 10mm Roof Beam Rebar                    |             | 393.96  | M     |
| <b>FORMWORKS</b>                        |             |         |       |
| TOTAL HEIGHT                            |             | 92      | M     |
| COLUMN FORMWORKS                        |             | 110.4   | SQ M  |
| TOTAL LENGTH                            |             | 87.28   | M     |
| SUSPENDED BEAM FORMWORKS                |             | 87.28   | SQ M  |
| SLAB FORMWORKS                          |             | 64.23   | SQ M  |
| <b>CEILING WORKS</b>                    |             |         |       |
| TOTAL PERIMETER                         |             | 235.36  | M     |
| 3.5mm THK Ficem Board                   |             | 122.19  | SQ M  |
| PVC SPANDREL                            |             | 41.01   | SQ M  |
| <b>TILE WORKS</b>                       |             |         |       |
| 0.3x0.3 Non-skid Ceramic Tiles          |             | 14.71   | SQ M  |
| 0.3x0.3 Non-skid Textured Ceramic Tiles |             | 2.52    | SQ M  |
| 0.4x0.4 Ceramic Tiles                   |             | 53.7    | SQ M  |
| 0.4x0.4 Non-skid Ceramic Tiles          |             | 8.81    | SQ M  |
| 0.6x0.6 Ceramic Tiles                   |             | 50.66   | SQ M  |
| Plain Cement Groove Finish              |             | 41.63   | SQ M  |
| Solid Wood Plank                        |             | 5.74    | SQ M  |

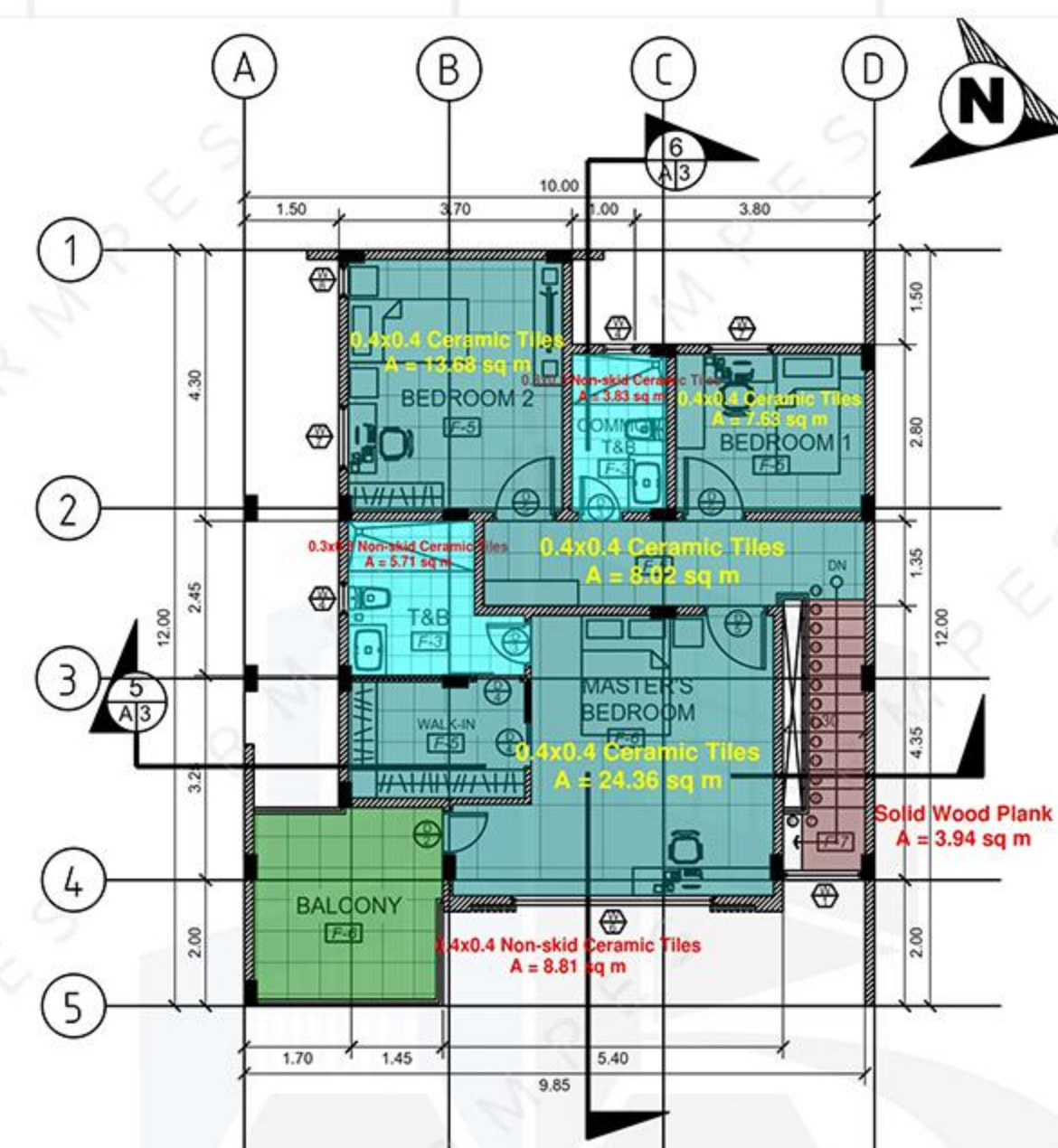


## SAMPLE BLUEBEAM REVU QUANTITY TAKEOFF

### 1 TILE WORKS



1 GROUND FLOOR PLAN  
A-2 SCALE 1:75M



2 SECOND FLOOR PLAN  
A-2 SCALE 1:75M

#### FLOOR FINISHES

| Description                             | Quantity | Unit |
|-----------------------------------------|----------|------|
| 0.3x0.3 Non-skid Ceramic Tiles          | 14.71    | sq m |
| 0.3x0.3 Non-skid Textured Ceramic Tiles | 2.52     | sq m |
| 0.4x0.4 Ceramic Tiles                   | 53.70    | sq m |
| 0.4x0.4 Non-skid Ceramic Tiles          | 8.81     | sq m |
| 0.6x0.6 Ceramic Tiles                   | 50.66    | sq m |
| Plain Cement Groove Finish              | 41.63    | sq m |
| Solid Wood Plank                        | 5.74     | sq m |

### 2 CEILING WORKS

#### CEILING FINISHES

| Description | Label                             | Quantity | Unit | Length |
|-------------|-----------------------------------|----------|------|--------|
| C-1         | 3.5mm Ficemboard on Metal Furring | 122.19   | sq m | 150.25 |
| C-2         | PVC SPANDREL                      | 41.01    | sq m | 85.11  |

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# 04 SAMPLE ESTIMATING PROJECTS

## 4. CONSTRUCTION OF 2 KM ROAD – BRGY. KALIKID SUR, CABANATUAN CITY, NUEVA ECIJA

Estimated Construction Cost: USD 207,351.00

Scope: Road Plan & Profile | Civil 3D | DUPA | BOQ/Program of Works | DPWH Standards.

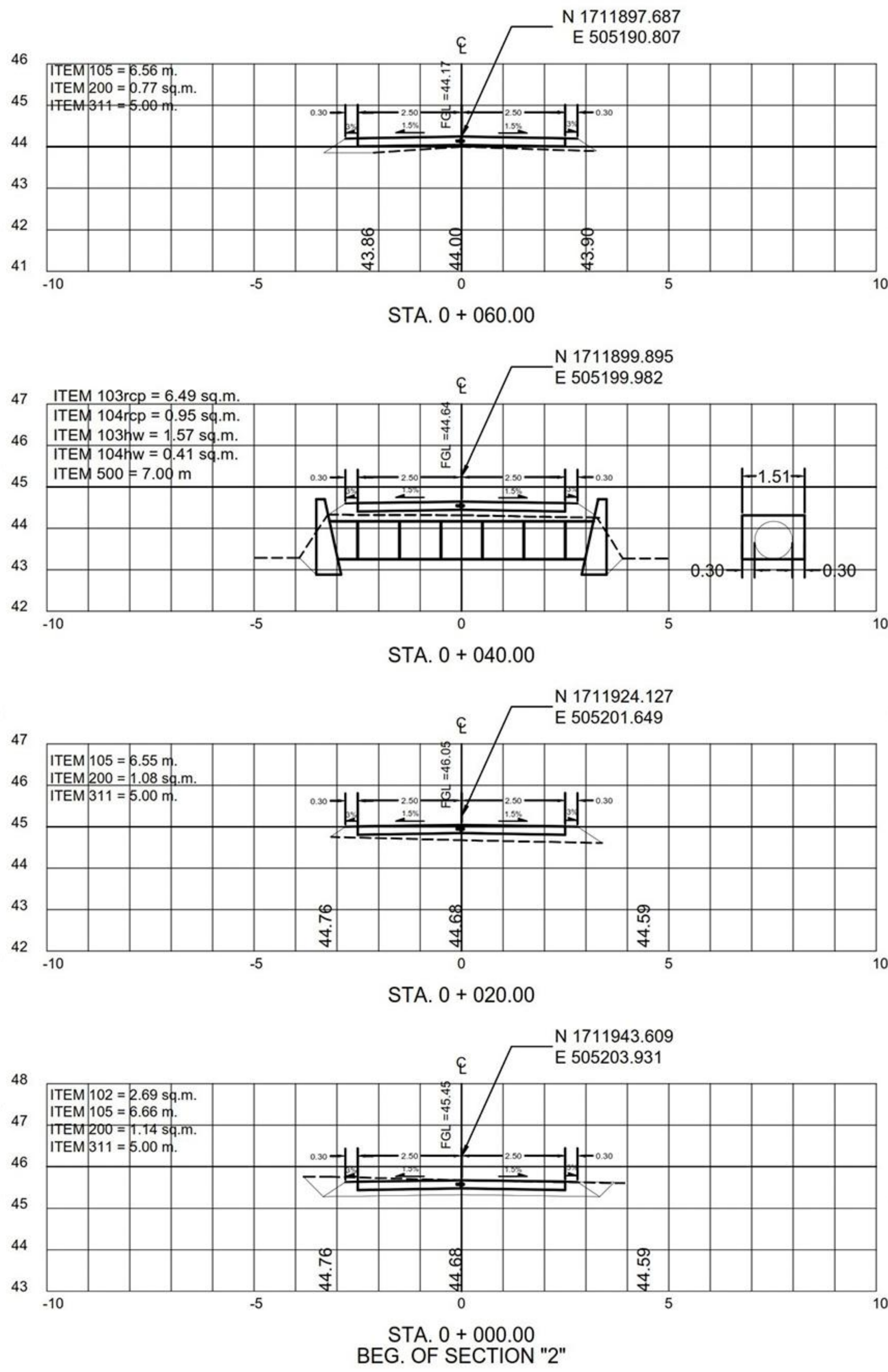
### SAMPLE PROGRAM OF WORKS | BOQ

| DESCRIPTION OF WORK TO BE DONE           | % of Total | EQUIPMENT                |        |           |
|------------------------------------------|------------|--------------------------|--------|-----------|
|                                          |            | DESCRIPTION              | NEEDED | AVAILABLE |
| FACILITIES FOR THE ENGINEER              | 0.47%      | Motorized Road Grader    | 1      |           |
| OTHER GENERAL REQUIREMENTS               | 2.85%      | Vibratory Roller         | 1      |           |
| EARTHWORKS                               | 2.54%      | Water Truck/Pump         | 1      |           |
| SUBBASE AND BASE COURSE                  | 33.06%     | One Bagger Mixer         | 1      |           |
| SURFACE COURSE                           | 59.63%     | Concrete Vibrator        | 1      |           |
| BRIDGE CONSTRUCTION                      | 0.60%      | Bar Cutter, Single Phase | 1      |           |
| DRAINAGE AND SLOPE PROTECTION STRUCTURES | 0.85%      | Concrete Saw             | 1      |           |
|                                          | 100.00%    |                          |        |           |

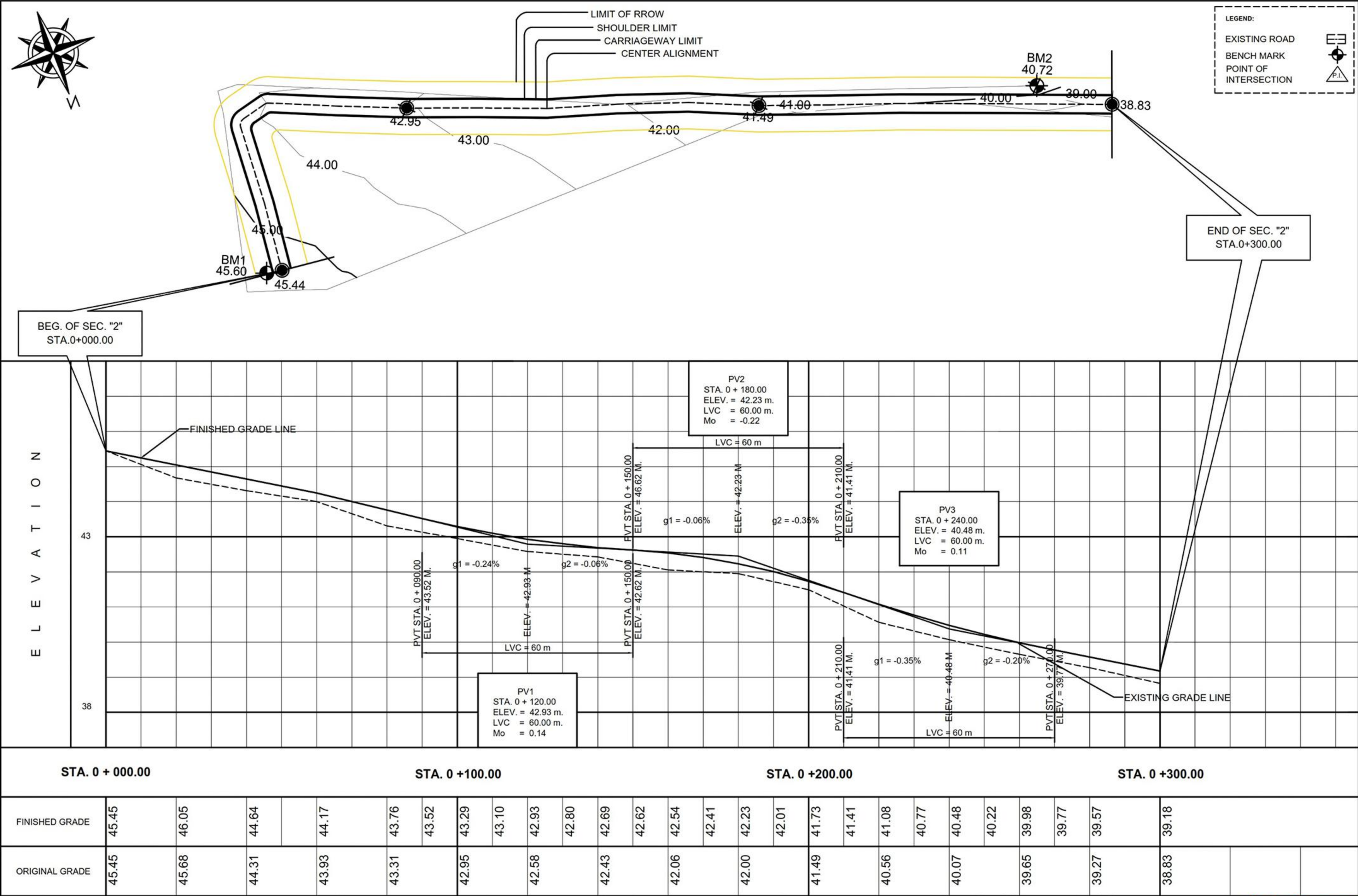
| Spec's Item No.                                        | DESCRIPTION                                                      | Unit | Quantity | DIRECT COST |              |
|--------------------------------------------------------|------------------------------------------------------------------|------|----------|-------------|--------------|
|                                                        |                                                                  |      |          | Unit Cost   | Total Cost   |
| <b>PART A FACILITIES FOR THE ENGINEER</b>              |                                                                  |      |          |             |              |
| A.1.1(8)                                               | Provision of Field Office for the Engineer (Rental Basis)        | mo.  | 3.00     | 17,325.00   | 51,975.00    |
| <b>PART B OTHER GENERAL REQUIREMENTS</b>               |                                                                  |      |          |             |              |
| B.5                                                    | Project Bill board/ Signboard                                    | ea   | 3.00     | 6,392.34    | 19,177.02    |
| B.7(2)                                                 | Occupational Safety & Health Program                             | ls.  | 1.00     | 206,936.31  | 206,936.31   |
| B.8(1)                                                 | Traffic Management                                               | mo.  | 3.00     | 2,611.22    | 7,833.66     |
| B.9                                                    | Mobilization/Demobilization                                      | ls.  | 1.00     | 81,656.47   | 81,656.47    |
| <b>PART C EARTHWORKS</b>                               |                                                                  |      |          |             |              |
| 101(4)a1                                               | Removal of Actual Structures/Obstruction (610mm dia.)(RCPC)      | l.m. | 16.00    | 506.59      | 8,105.38     |
| 102(2)                                                 | Surplus Common Excavation                                        | m³   | 150.20   | 188.79      | 28,356.26    |
| 103(1)a                                                | Structure Excavation (Common Soil)                               | m³   | 22.40    | 309.62      | 6,935.49     |
| 104(1)a                                                | Embankment From Roadway Excavation                               | m³   | 120.16   | 128.02      | 15,382.53    |
| 104(2)a                                                | Embankment From Borrow, Common Soil                              | m³   | 179.31   | 685.83      | 122,976.18   |
| 105(1)a                                                | Subgrade Preparation (Common Material)                           | m²   | 4,660.00 | 21.34       | 99,444.40    |
| <b>PART D SUBBASE AND BASE COURSE</b>                  |                                                                  |      |          |             |              |
| 200(1)                                                 | Aggregate Subbase Course                                         | m³   | 3,824.85 | 958.17      | 3,664,856.52 |
| <b>PART E SURFACE COURSE</b>                           |                                                                  |      |          |             |              |
| 311(1)b1                                               | Portland Cement Concrete Pavement (Unreinforced) ( 0.20m thick), | m²   | 5,000.00 | 1,322.04    | 6,610,208.86 |
| <b>PART F BRIDGE CONSTRUCTION</b>                      |                                                                  |      |          |             |              |
| 404(1)a                                                | Reinforcing Steel Bar, Grade 40 (Minor Structures)               | kgs. | 28.14    | 81.38       | 2,290.04     |
| 405(1)a3                                               | Structural Concrete Class A (Minor Structures)                   | m³   | 8.06     | 8,001.18    | 64,489.55    |
| <b>PART G DRAINAGE AND SLOPE PROTECTION STRUCTURES</b> |                                                                  |      |          |             |              |
| 500(1)b1                                               | Pipe Culverts (610mm dia.), Class IV                             | l.m. | 14.00    | 3,444.73    | 48,226.27    |
| 500(1)b3                                               | Pipe Culverts (910mm dia.), Class IV                             | l.m. | 7.00     | 6,586.31    | 46,104.18    |

|                                                   |            |               |              |        |                                               |
|---------------------------------------------------|------------|---------------|--------------|--------|-----------------------------------------------|
| BREAKDOWN OF ESTIMATED EXPENDITURES :             |            | % of Total    |              |        |                                               |
| Labor :                                           | Man-days : | P             | 1,235,679.44 | 11.04% | A. Direct Cost Sub-Total : P 11,084,954.12    |
| Materials :                                       | P          | 6,398,742.41  | 57.15%       |        | B. Construction Eng'g. : 112,000.00           |
| Rental of Equipment :                             | P          | 854,524.86    | 7.63%        |        | C. Total & Miscellaneous :                    |
| Fuel,Oil,Spare Parts & Maintenance of Equipment : |            |               |              |        | D. Contingencies :                            |
| Others :                                          | P          | 2,596,007.41  | 23.18%       |        | E. Prelim, Engineering :                      |
| Overhead :                                        | P          | 112,000.00    | 1.00%        |        | F. Quality Control :                          |
| TOTAL :                                           | P          | 11,196,954.12 | 100.00%      |        | G. Project Supervision : P 11,196,954.12      |
|                                                   |            |               |              |        | H. Total Estimated Cost : SAY P 11,200,000.00 |

### 1 Road Cross Section



### 2 Road Plan and Profile



Note: Drawings and project information are presented for training and professional portfolio purposes. Actual project records and confidential information are not reproduced.

# 05 CONSTRUCTION & QA/QC EXPERIENCE

## 1. 500 KV TRANSMISSION LINE GRID INTERCONNECTION – TERRA SOLAR PROJECT

**PROJECT SCOPE:** Earthworks • Site Layout & Survey • Tower Foundations • Reinforcement & Formwork • Concrete Works • Tower Erection • Material Inspection

### ROLES AND RESPONSIBILITIES:

- Reviewed construction drawings, rebar details, **Bar Bending Schedules (BBS)**, rebar cutting lists, **BOQ**, and **BOM** to support construction accuracy and quantity requirements.
- Monitored tower foundation and erection activities, including **progress monitoring**, **site inspections**, inspection reports, and construction documentation.
- Conducted and documented **material inspections and testing** in accordance with applicable ACI and ASTM requirements, including concrete compressive strength and rebar tensile and bending tests.
- Prepared and monitored **RFIs**, **Inspection and Test Plan (ITP)**, **progress billing documents**, BOQ/BOM reports, **Job Order Requests (JOR)**, and **Matrix of Responsibility (MOR)** to support project implementation and documentation.
- Conducted material receiving inspections and supported warehouse management, material scheduling, and inventory documentation.
- Coordinated site activities and **technical meetings** with MTerra Solar Philippines, JACOBS, Energy China, Power China, subcontractors, and liaison personnel, supporting project progress and **ISO 9001-based QA/QC requirements**.

### TRANSMISSION TOWER CONSTRUCTION METHODOLOGY

#### FOUNDATION WORKS

1

##### Site Preparation:

Site clearing, leveling, grading and survey works layout.

2

##### Excavation:

Excavate to the design depth and dimensions; ensure safe slopes/shoring.

3

##### Setting out and Alignment:

Set the foundation center and check position, level, and alignment.

4

##### Reinforcement & Anchor bolt Installation:

Install as per design and ACI Standard.

5

##### Formworks Installation:

Install formworks with proper support and bracing.

6

##### Concrete Pouring/Casting:

Pour concrete as per approved mix design and ensure proper consolidation.

7

##### Concrete Curing:

Allow concrete to set for at least 24 hrs. Cure the concrete for required period.

8

##### Backfilling:

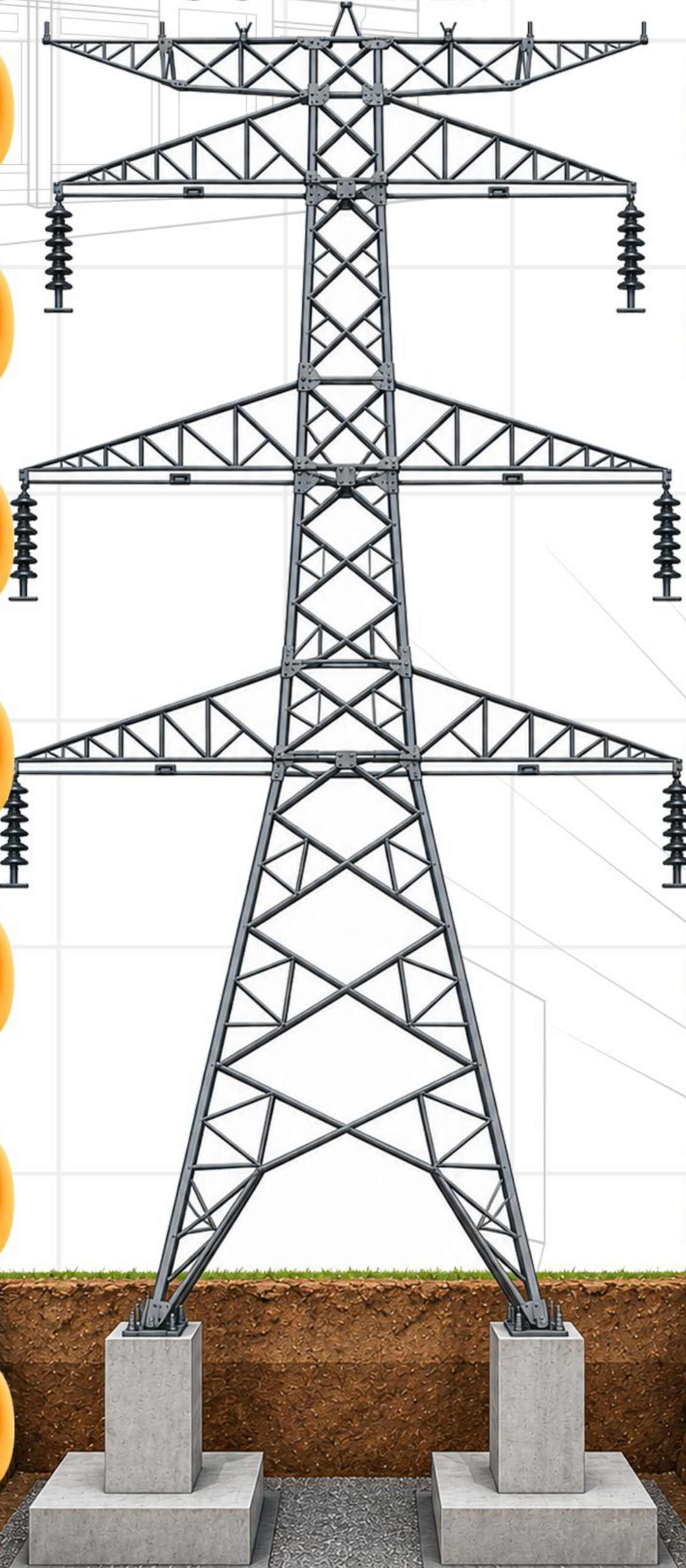
Backfill with suitable material and compact properly.

9

##### Quality Control:

Inspect reinforcement, anchor bolts, formwork, dimensions, and concrete quality.

#### 90 TOWERS



#### ERECTION WORKS

10

##### Materials & Equipment:

Prepare and inspect all tower components, lifting equipment and accessories.

11

##### Site Preparation:

Clear the site of any debris or obstacle provide sufficient working space and access route.

12

##### Tower Component Assembly:

Sort and Assemble tower members at laydown area as per drawings.

13

##### Lifting and Erection:

Lift and install tower section using appropriate cranes and lifting procedures.

14

##### Alignment and Positioning:

Inspect the tower vertically, ensure the alignment of members according to drawings.

15

##### Bolt Tightening:

Install and tighten bolts to the specified torque. Apply locknuts and lock washers.

16

##### Inspection:

Conduct thorough inspection of all members, connection and accessories.

17

##### Quality Control:

Ensure all works comply with drawings, specification and applicable standards.

**Project Standards:**  
**Foundation Works**  
ACI 318 | ACI PRC-315 | ASTM C143 |  
ASTM C31 | ASTM C39 | ASTM C172  
**Erection Works**  
ASCE 10 | AISC 360 | ASTM A123 |  
ASTM A394 | ASTM F3125 | IEC 60826

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# 05 CONSTRUCTION & QA/QC EXPERIENCE

## 2. MTERRA SOLAR PHASE 1 PHOTOVOLTAIC PROJECT – 1.4 GW SOLAR PV

**PROJECT SCOPE:** Site Clearing & Grading • Site Layout • Screw Pile Installation  
Mounting Structures • PV Module Installation

### ROLES AND RESPONSIBILITIES:

- Supervised **civil and structural construction activities**, conducted **quality trainings** on proper construction methodologies from screw pile installation to mounting structure and PV module installation, ensuring compliance with approved drawings and specifications.
- Conducted **QA/QC inspections** in accordance with **ISO 9001**, including construction activities inspections, material verification, testing, workmanship monitoring, and quality control.
- Prepared and managed **Work Inspection Request (WIR)**, **Quality Control Reports (QCRs)**, **Non-Conformance Reports (NCRs)**, inspection records, quality certifications, and other **QA/QC documentation** through project closeout.
- Monitored project **progress and scheduling**, coordinated site activities, and followed up on work targets and pending works.
- Coordinated with MTerra Solar Philippines, JACOBS, CEEC, subcontractors, and suppliers to facilitate inspections, resolve technical issues, and maintain project progress.

## PHOTOVOLTAIC CONSTRUCTION METHODOLOGY

### SCREW PILE INSTALLATION

1

**Site Preparation:**  
Site clearing, leveling,  
grading and survey works  
layout.

2

**Staked Point Layout:**  
Verify the location, spacing, and  
alignment of staked points against  
the approved plan before piling.

3

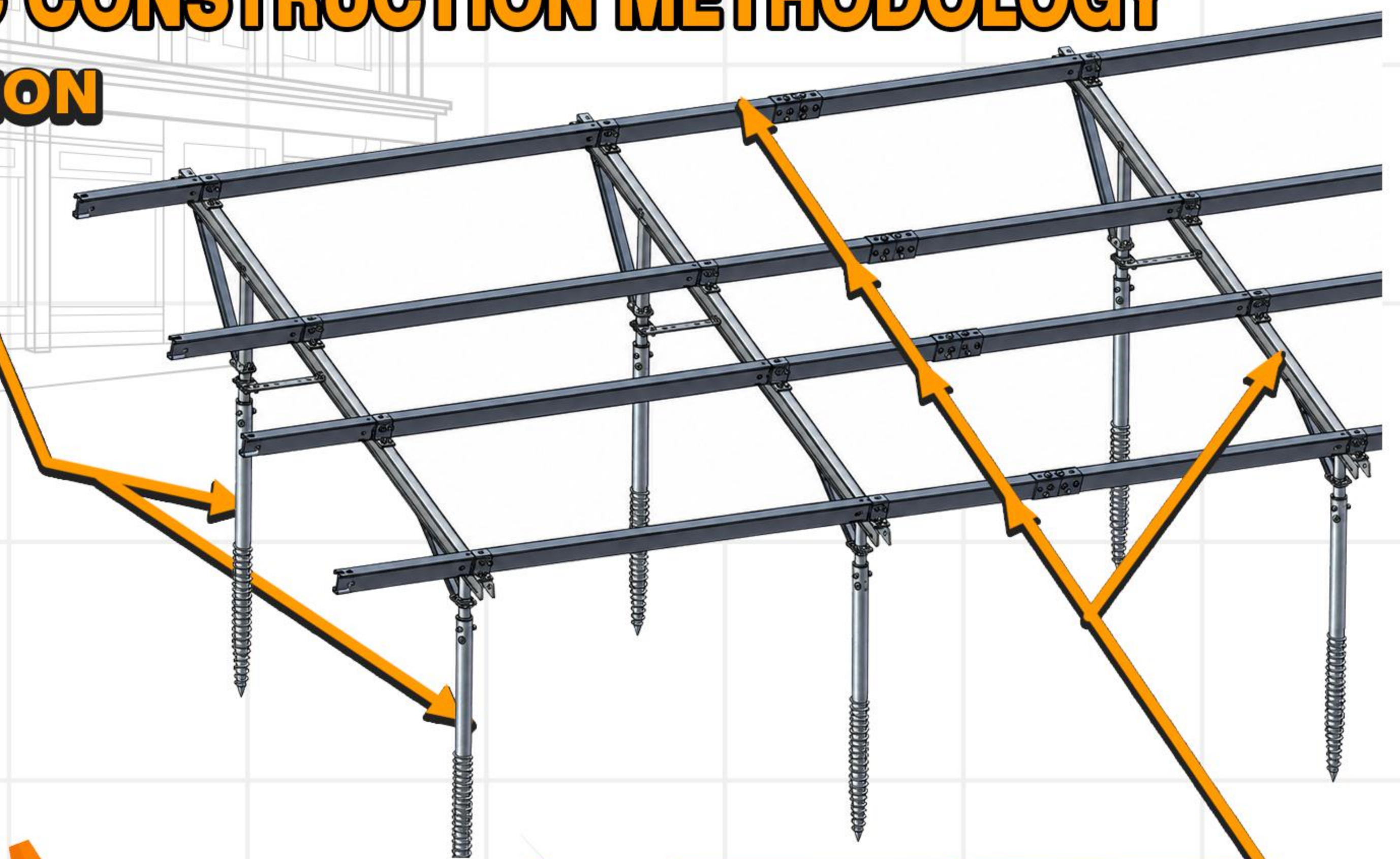
**Ground Screw Piling:**  
Verify the location, elevation,  
and plumbness of screw piles  
based on approved plans.

4

**Screw Pile Rectification:**  
Monitor the rectification of out  
of tolerance, refusal, and  
damage piles.

5

**QA/QC Inspection:**  
Inspect materials, equipment, &  
installation, for compliance with  
the approved methodologies.



### MOUNTING STRUCTURE INSTALLATION

6

**Material Inspection:**  
Organized and inspect  
materials for damage, defects,  
and completeness.

7

**Main Rail & Strut Installation:**  
Installed posts and struts at each piles,  
connect struts at main rails, and verified  
proper bolt installation and torque.

8

**Subrail & Tension Rod Installation:**  
Installed subrails in the specified  
sequence and ensured proper  
installation of tension rods.

9

**QA/QC Inspection:**  
Inspection of structure in accordance  
to the plans and specification, verified  
bolt torque and member completeness.

### PV MODULE INSTALLATION

10

**Module Installation:** Installed modules  
by verifying proper clamp connection,  
alignment, and positioning. Followed the  
specified installation sequence.

11

**QA/QC Inspection:** Verified module  
installation in accordance to plan.  
Inspect modules for construction  
related damage and ensured proper  
rectification.

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# 05 CONSTRUCTION & QA/QC EXPERIENCE

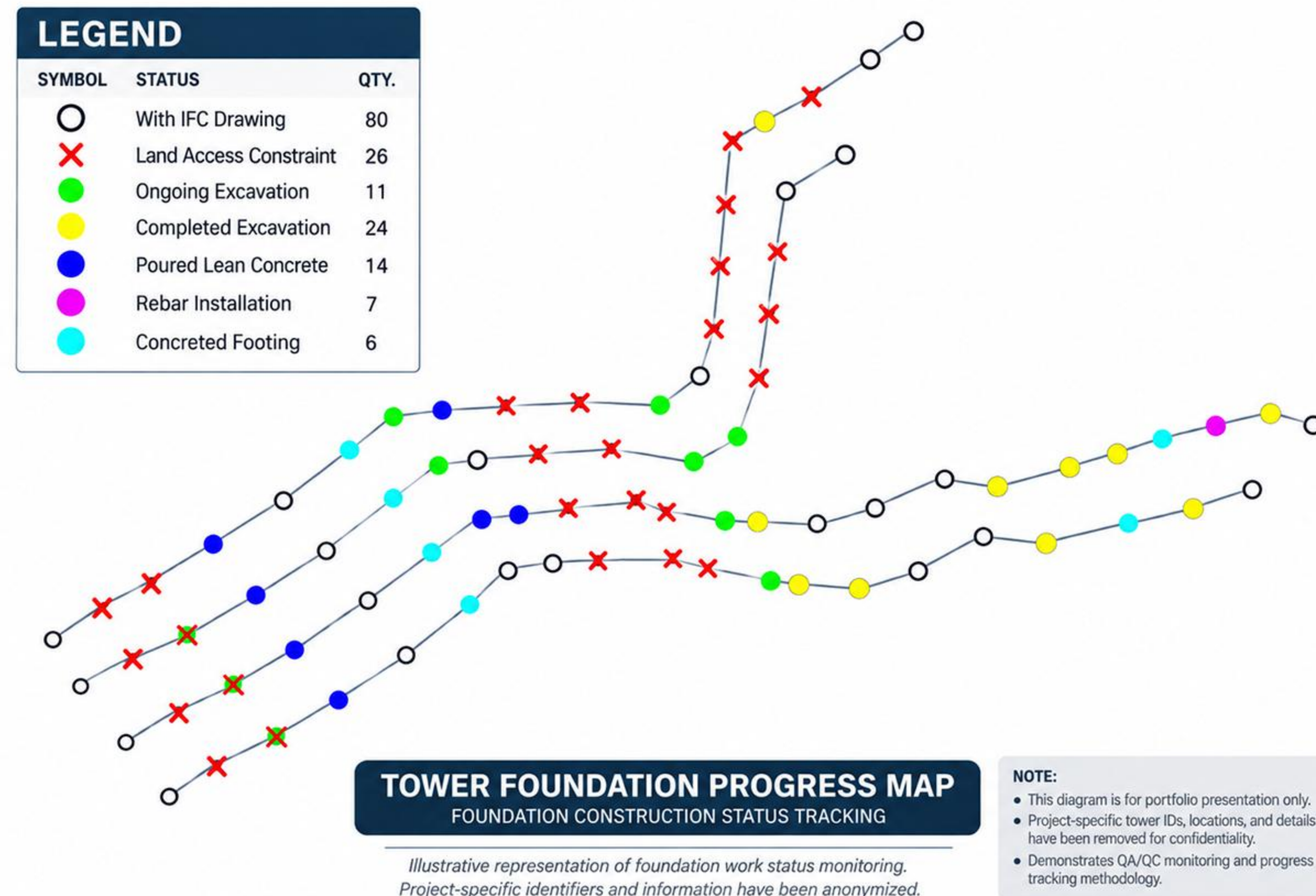
## 3. 35 KV OVERHEAD LINE TOWER – TERRA SOLAR POWER PLANT

**PROJECT SCOPE:** Earthworks • Site Layout Survey • Tower Substructure/Foundation • Tower Erection

### ROLES AND RESPONSIBILITIES:

- Monitored tower construction progress and site activities, including earthworks, tower foundation, anchor bolt installation, concrete works, and tower erection.
- Prepared and scheduled **WIRs**, coordinated inspections, and monitored **QCR/NCR** status, **rectification works**, and close-out documentation.
- Prepared **Work Method Statements (WMS)** for earthworks, tower foundation, and tower erection, ensuring proper construction methodologies and QA/QC requirements.
- Conducted **site inspections** and **coordinated** with subcontractors and project stakeholders to address technical issues, implement rectifications, and maintain quality and construction progress.

### TOWER PROGRESS MONITORING 80 TOWERS

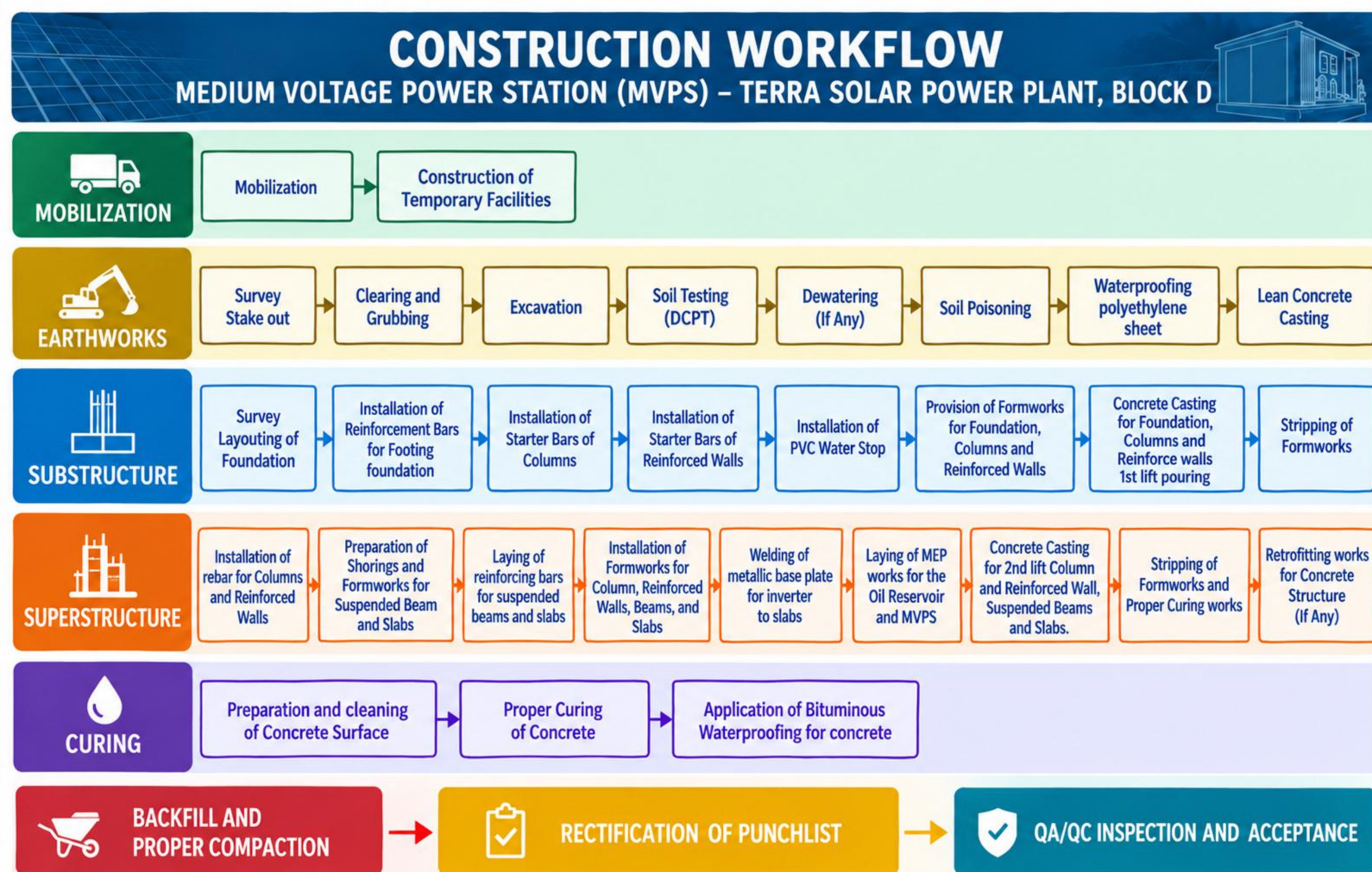


## 4. MEDIUM VOLTAGE POWER STATION (MVPS) – TERRA SOLAR POWER PLANT, BLOCK D

**PROJECT SCOPE:** Earthworks • Site Layout Survey • Substructure • Superstructure

### ROLES AND RESPONSIBILITIES:

- Inspected **concrete pouring** for substructure and superstructure works in accordance with **ACI 304R-00**, ensuring proper placement, consolidation, and curing.
- Performed and inspected **fresh concrete testing and sampling**, including Slump (**ASTM C143**), Temperature (**ASTM C1064**), Specimen Making & Curing (**ASTM C31**), and Compressive Strength (**ASTM C39**).
- Verified concrete quality and design mix in accordance with **ACI 211.1**, ensuring required strength, durability, and workability.



## 5. CONTROL BUILDING – TERRA SOLAR POWER PLANT, BLOCK D

**PROJECT SCOPE:** Earthworks • Site Layout Survey • Substructure

### ROLES AND RESPONSIBILITIES:

- Performed **site layout** and **provided survey data** for foundation excavation, rebar, concrete layout, and elevations in accordance with approved design drawings.
- Inspected and monitored **foundation works**, including **Dynamic Cone Penetration Test (DCPT)** soil testing and bearing capacity verification, as well as **concrete and rebar testing** in accordance with applicable standards.
- Prepared and managed **WIRs, ITPs, rectification works, and site progress documentation**, ensuring QA/QC compliance and proper close-out.

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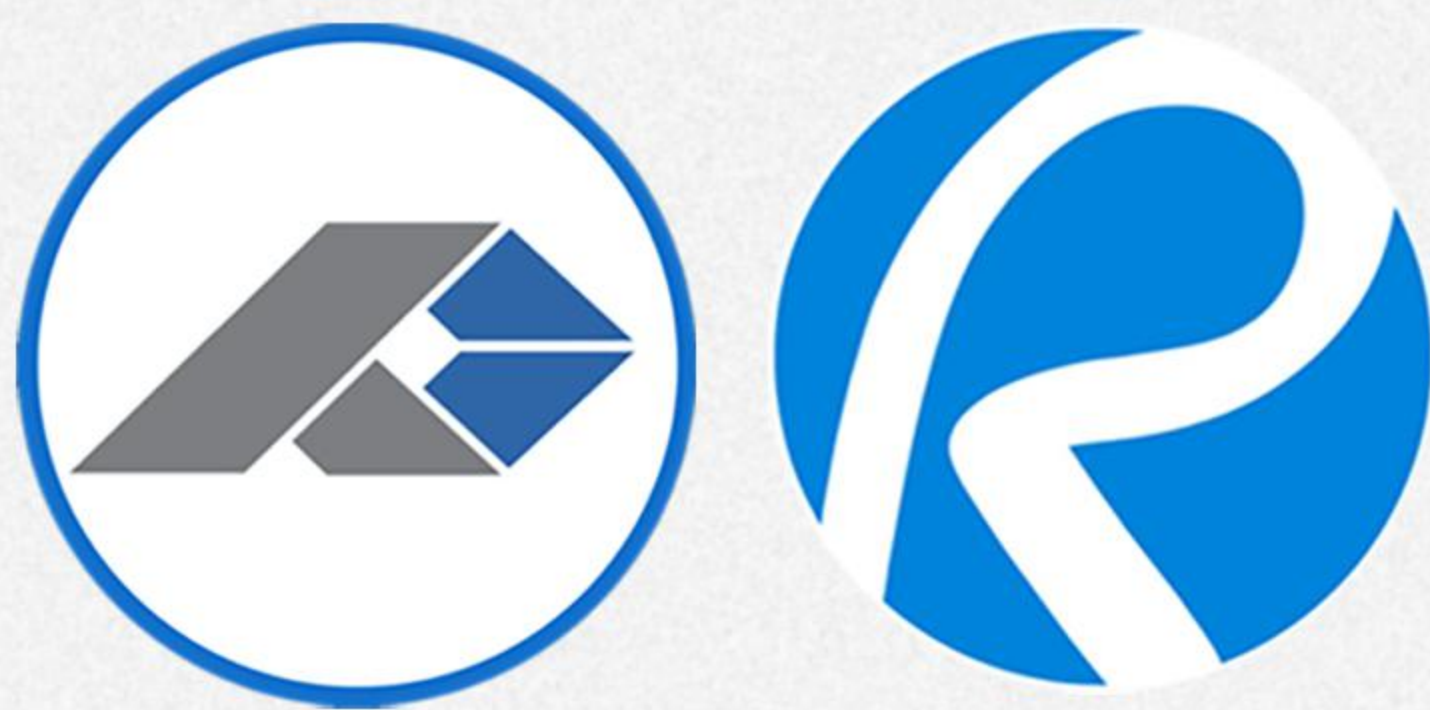
# 06

# CERTIFICATIONS & TRAINING

## 1. BLUEBEAM AND PLANSWIFT TRAINING AND WORK SHOP: DIGITAL QUANTITY TAKEOFF AND ESTIMATION FOR AEC PROFESSIONALS

**XSTRUCTURES ENGINEERING CONSULTANTS**  
**AUGUST 3 – 7, 2026**

**COURSE OUTLINE:** User Interface | Plan Management | Plan Overlay & Comparison | Annotations & Scaling | Takeoff Tools | Organization & Visual Control | Cost Estimation | Template & Reporting Workflows | Software



## 2. MANUAL CONSTRUCTION BUILDING ESTIMATE: WITH AID OF MICROSOFT EXCEL AND AUTOCAD



**MSTCONNECT | AUGUST 12 – 16, 2024**

**COURSE OUTLINE:** Introduction to Estimate | Structural Estimate | Architectural Estimate | Sanitary & Electrical | Cost Estimate in Actual



## 3. QUALITY ASSURANCE & QUALITY CONTROL IN CONSTRUCTION

**MSTCONNECT | JULY 11, 2025**

**COURSE OUTLINE:** Fundamentals of QA/QC | Industry Standards & Code Compliance | QA/QC Planning | Site Deficiency Prevention | Documentation & Auditing | Quality Monitoring





# 06

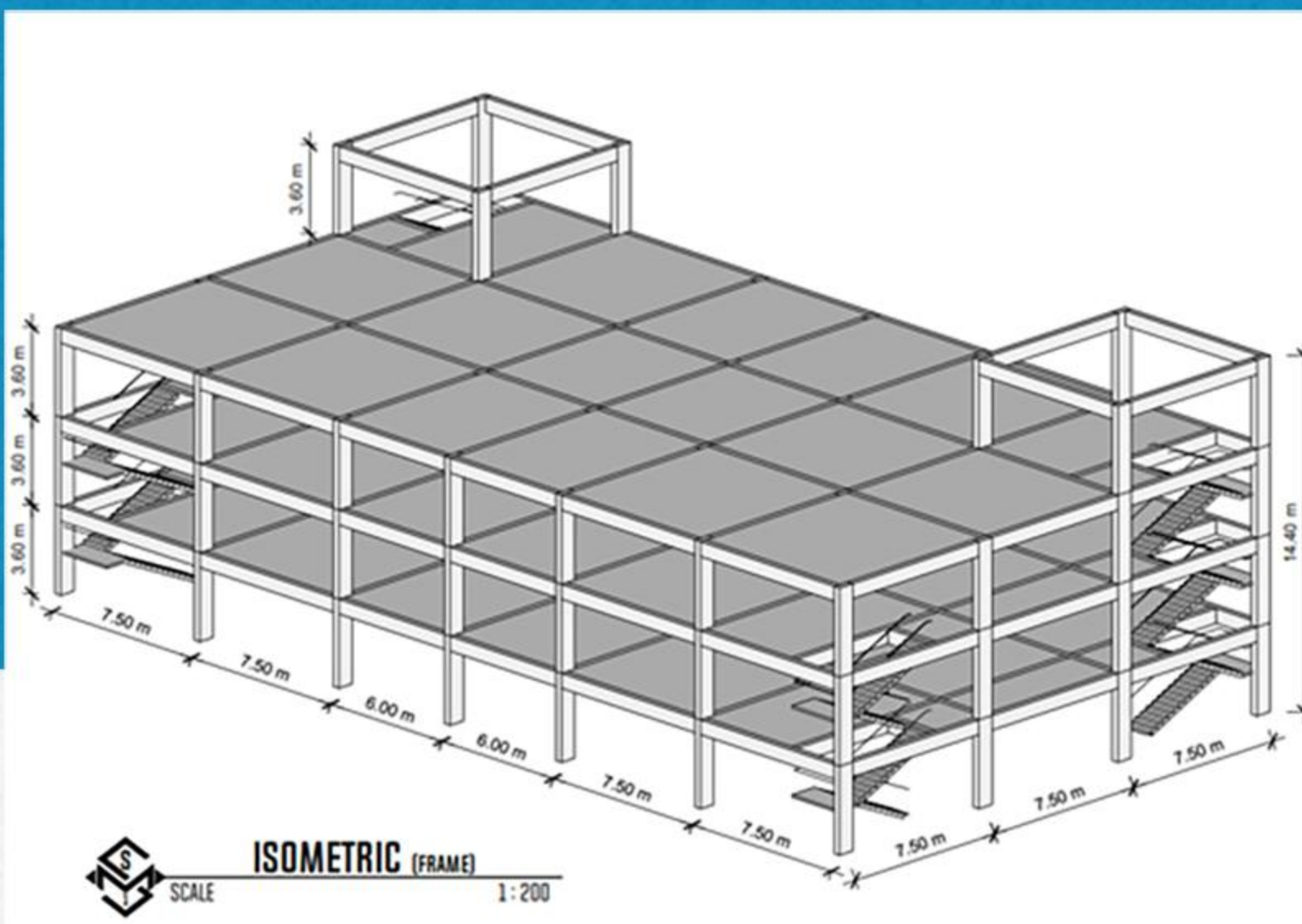
# CERTIFICATIONS & TRAINING

## 4. STRUCTURAL ANALYSIS & DESIGN — 3-STOREY COMMERCIAL BUILDING | NSCP 2015 & STAAD

PHILIPPINE INSTITUTE OF CIVIL ENGINEERS,  
CENTRAL LUZON STATE UNIVERSITY  
MAY 27, 2023



STAAD.Pro



## 5. CERTIFICATE OF COMPLETION | DPWH ENGINEERING APPRENTICESHIP



DPWH PLANNING & DESIGN SECTION  
JUNE 14, 2022 TO JULY 23, 2022

SCOPE: Planning, Design & Estimating of  
Horizontal and Vertical Structures

# THANK YOU. LET'S CONNECT

AVAILABLE FOR: Remote / WFH,  
Construction Estimating, Quantity Takeoff,  
Quantity Surveying, Construction VA



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